

AE-340: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNN})]$ via $[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$

Date: 2024-09-02

Tags: PNN $[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$
Dichloro NMR AE ^1H ^{31}P

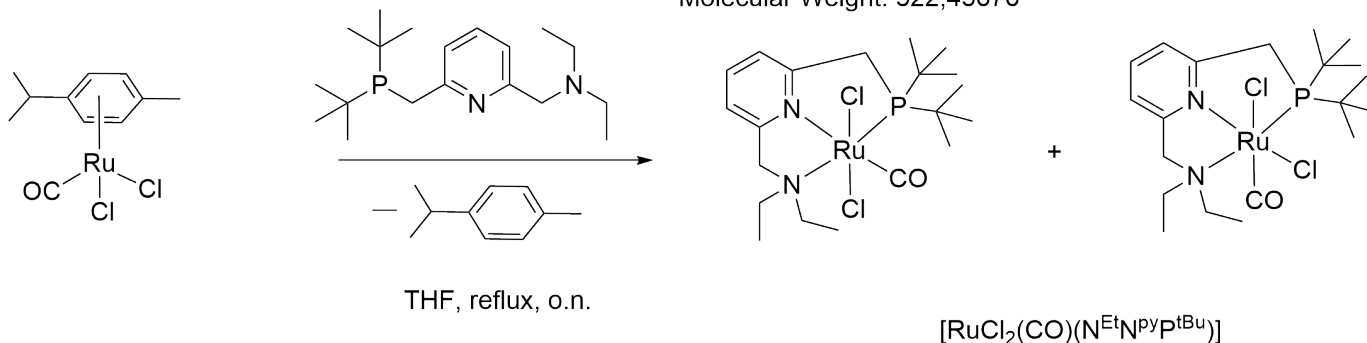
Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure

Chemical Formula: $\text{C}_{20}\text{H}_{35}\text{Cl}_2\text{N}_2\text{OPRu}$

Molecular Weight: 522,45676



$[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$ $\text{N}-((6-((\text{di-tert-butylphosphane})\text{methyl})\text{pyridin-2-yl})\text{methyl})\text{-N-ethylethanamine}$
Chemical Formula: $\text{C}_{19}\text{H}_{35}\text{N}_2\text{P}$
Molecular Weight: 322,47676

Molecular Weight:
334.20 g/mol

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Research report AE Experiment - AE-232: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$ AE-261: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$

Reagents

Name	CAS Number/ Experiment number	Amount [mmol]	Equivalents	Purity	Mass _{theo,pure} [mg]	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
$[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$	AE-221: Synthesis of $[\text{RuCl}_2(\text{CO})(p\text{-cymene})]$	0.375	1.00	0.58	125.57	216.50	216.50	334.20	/
PNN, 6-[[Bis(1,1-dimethylethyl)phosphino)methyl]-N,N-diethyl-2-pyridinemethanamine	863971-66-8	0.747	2.00	/	/	243.79	241	322.47	/
THF	AE-289: Degassing of THF	/	/	/	/	/	/	/	14
Et2O	JU-KRA-172: Degassing of Diethylether	/	/	/	/	/	/	/	15

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an nitrogen filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Date	Time	Step	Observations
02.09		A 50 mL and a 20 mL Schlenk flask were prepared	
	14:00 - 15:00	The air cooler was dried by setting under vacuum	approx. 1:00 h
	14:15	[Ru] was added to the 50 mL flask	red solid Ru.jpg Ru suspension.jpg
	14:50	In the glovebox in K004 PNN was added to the 20 mL flask	yellow/orange liquid PNN.jpg PNN solution.jpg
	15:15	To the [Ru] THF (5.5 mL) was added, to the PNN THF (8.5 mL) was added	Ru: red suspension, PNN: slightly yellow solution
	15:25	The PNN solution was added to the [Ru] suspension	red suspension
		The reaction mixture was stirred at rt and the flask was connected to the cooler in argon counterflow.	
	15:30 - 15:50	The reaction mixture was heated to reflux	before heating.jpg
	15:50	The reaction mixture was heated at reflux o.n.	
03.09	8:15	The heating was turned off and the mixture was allowed to rt.	at rt.jpg
	8:50 - 9:00	The volume of the remaining reaction mixture was reduced under reduced pressure to approx. half	reduced volume.jpg
	9:10	To the solution Et ₂ O (15 mL) was added under stirring.	after addition of Et₂O.jpg
		The mixture was stored under argon at -18 °C	
04.09	10:40 - 11:05	The obtained mixture was filtered according to Filtration with frit technique and the residue was washed with n-Heptane (3 * 5 mL)	before filt.jpg
	11:10 - 12:30	The obtained solid was dried under reduced pressure.	AE-340-1

	12:45	A NMR-sample was prepared from AE-340-1 according to [Protocol] Preparation of NMR Sample	AE-340-1
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Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative data	Interpretation
04.09	09:30	AE-340-1	¹ H, ³¹ P (quant)	500 MHz IOMC	DCM-d2	AE-340-1.zip	AE-340-1_10.nmrium	Research report AE, p.30	Desired product obtained. 95 % pure. 85 % cis species, 15 % trans species

Product characterization

	mass [mg]	purity [%]	mass _{pure} [mg]	amount [mmol]	Yield [%]	description	storage location
AE-340-1	120.82	95	114.78	0.217	58.5	ochre solid	empty

Linked experiments

- [JSC-KS-04: Degassing of n-Heptane](#)

- [AEI-047: Synthesis of 2,6-Bis\(P,P-di\(tertbutyl\)phosphinomethyl\)pyridine](#)

- [AEI-057: Synthesis of \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AEI-060: Synthesis of \[RuCl₂\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AEI-072: Synthesis of \[RuCl₂\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AE-100: Synthesis of \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AE-101: Synthesis of \[RuCl₂\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AE-102: Degassing of THF](#)

- [AE-108: Synthesis of \[RuCl₂\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- [AE-120: Degassing of n-Heptane](#)

- [AE-157: Synthesis of \[RuCl₂\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl₂\(CO\)\(p-cymene\)\]](#)

- KRA-080: Degassing of THF

- AE-221: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-230: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-232: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-248: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-253: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine

- AE-261: Synthesis of $[\text{RuCl}_2(\text{CO})(\text{PNP})]$ with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$

- AE-289: Degassing of THF

- AE-315: Degassing of n-Heptane

- JU-KRA-172: Degassing of Diethylether

Linked resources

Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

Protocol - [Schlenk Technique](#)

Attached files

20240904_134457.jpg

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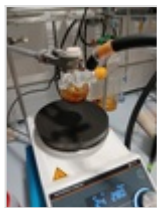
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at-rt.jpg

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PNN.jpg

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Ru.jpg

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before-heating.jpg

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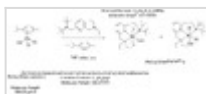
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AE-340.png

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AE-340.cdxml

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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1321>